

School of Arts and Sciences
Department of Computer Science/Information Technology
Fall 2025-2026
CSCI345 Digital Logic

Class Day : _____ **Class Time** : _____ **Room**: _____
Instructor : _____
E-mail : _____
Office : _____ **Office Hours** : _____ **Extension No**: _____

I. Description

The course develops the ability of the student to understand the design of digital electronic circuits, which are the main components in digital computers, data communication, digital recording, etc. The course covers number systems, Boolean switching algebra, combinational circuit design, flip-flops, counters, registers, state machine notation, analysis of sequential circuits, and sequential circuit design.

II. Course Objectives:

- Employ Digital Logic Gates
- Learn Microcomputer Architecture
- Implement and construct various combinational and sequential digital circuits
- Investigate Number Systems and Boolean Algebra

III. Course Learning Outcomes:

CLO-1: Convert between the four major number systems: Binary, Octal, Decimal, and Hexadecimal (Add/Subtract binary numbers).

CLO-2: Solve and reduce Boolean functions.

CLO-3: Analyze and design basic combinational circuits using logic gates or multiplexers, or decoders.

CLO-4: Analyze sequential circuits and Design simple finite state machines from a specification and implement this using logic gates and edge-triggered flip-flops.

CLO-5: Choose the appropriate design and produce model circuits using LogiSim software.

IV. Mapping to program outcomes (*PLOs are listed at the end of the syllabus*)

Digital Logic	PLO - 1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1	X								
CLO-2	X								
CLO-3	X								
CLO-4		X							
CLO-5		X							

V. Topics Covered

- 1) Number System.
- 2) Boolean algebra.
- 3) Digital Logic Gates.
- 4) Combinational digital circuits.
- 5) Sequential digital circuits.

VI. Prerequisite – Corequisite

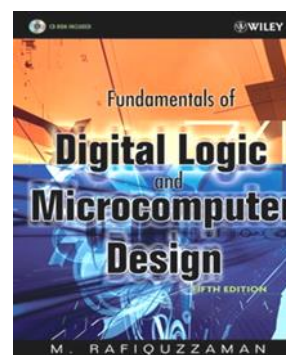
CSCI250: Introduction to Programming.

VII. Course Material:

Textbook: Fundamentals of Digital Logic and Microcomputer Design, by M. Rafiquzzaman, 5th edition, 2005
(ISBN: 0471727849) Publisher Wiley

Further Reading(s):

Textbook: Digital Design and Computer Architecture, by David Harris, 2nd Edition, 2012
(ISBN: 9789382291527) Publisher Wiley



VIII. Software(s) required

LogiSim

<http://www.cburch.com/logisim/index.html>

I. Course Assessment

Activity	Percentage	Date
Project	10	Jan 8, 2026, by midnight.
Test	20	Week 5, during second session
Midterm	35	Saturday, November 15, 2025, from 10:00 AM to 11:15 AM
Final Exam	35	TBA

II. Detailed Schedule

Week		Book Reference	Topic Details
Week 1 Sep 29	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 1: Introduction to Digital Systems <i>Briefly:</i> 1.1 Explanation of Terms, 1.4 Digital Integrated Circuits: Diodes, MOS Transistors
	Lecture 2		Chapter 2: Number Systems and Codes 2.1 Number Systems: Decimal, Binary, Octal, and Hexadecimal. Number Base Conversions
Week 2 Oct 6	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 2: Number Systems and Codes 2.2 Unsigned and Signed Binary Numbers
	Lecture 2		Chapter 2: Number Systems and Codes 2.3 Codes: BCD, ASCII, and briefly (EBCDIC, Excess-3, Gray, Unicode). 2.4 Fixed-Point and Floating-Point Representations
Week 3 Oct 13	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 2: Number Systems and Codes 2.5 Arithmetic operations with different bases, Binary Arithmetic (addition and subtraction), and BCD addition.
	Lecture 2		Chapter 3: Boolean Algebra and Digital Logic Gates 3.1 Basic Logic Operations (NOT, OR, AND) 3.2 Other Logic Operations (NOR, NAND, XOR, XNOR)
Week 4 Oct 20	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 3: Boolean Algebra and Digital Logic Gates 3.5 Boolean algebra: Truth Table, Boolean Identities, and simplification using Boolean Identities.
	Lecture 2		Chapter 3: Boolean Algebra and Digital Logic Gates 3.6 Standard Representations: Minterms and Maxterms, Sum of products (SOP) and product of sums (POS) expressions.
Week 5 Oct 27	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 3: Boolean Algebra and Digital Logic Gates 3.7 Karnaugh Maps: Two-, Three-, and Four-Variable K-Map.
	Lecture 2		Test during class time (Not common) from chapter 1 to 3.6
Week 6 Nov 3	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 3: Boolean Algebra and Digital Logic Gates 3.7 Karnaugh Maps: Prime Implicants, Expressing a function in POS form using K-Map.
	Lecture 2		Chapter 3: Boolean Algebra and Digital Logic Gates 3.7 Karnaugh Maps: Don't Care Condition, Minimum SOP form, Minimum POS form.

Week 7 Nov 10	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 3: Boolean Algebra and Digital Logic Gates 3.9 Implementation of Digital Circuits with NAND and NOR
	Lecture 2		Chapter 3: Boolean Algebra and Digital Logic Gates Problems solving and LogiSim simulation
			Midterm exam: Saturday, November 15, from 10:00 AM to 11:15 AM. It covers Chapter 2 to chapter 3.9
Week 8 Nov 17	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 4: Combinational Logic Design 4.1 Basic concepts. 4.2 Analysis of a combinational logic circuit.
	Lecture 2		4.3 Design of a Combinational Circuit, 4.4 Multiple-Output Combinational Circuit
Week 9 Nov 24	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 4: Combinational Logic Design 4.5 Typical Combinational Logic Circuits: Adders, Subtractors <u>LogiSim</u> : Designing an 8-bit Adder/ Subtractor, simulation.
	Lecture 2		Chapter 4: Combinational Logic Design 4.5 Typical Combinational Logic Circuits: Comparators, Decoders, Encoders
Week 10 Dec 1	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 4: Combinational Logic Design 4.5 Typical Combinational Logic Circuits: Multiplexers and Demultiplexers <u>LogiSim</u> : Designing a 7-segment display converter, simulation.
	Lecture 2		Problem Solving
Week 11 Dec 8	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 5: Sequential Logic Design 5.1 Basic Concepts, 5.2 Flip-Flops (RS, D, JK, and T), 5.5 Summary of Flip-Flops description.
	Lecture 2		Chapter 5: Sequential Logic Design Basic design steps
Week 12 Dec 15	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 5: Sequential Logic Design 5.6 Analysis of Synchronous Sequential Circuits.
	Lecture 2		Chapter 5: Sequential Logic Design 5.9 Design of Synchronous Sequential Circuits.
Week 13 Dec 22	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	Chapter 5: Sequential Logic Design 5.10 Design of Counters
	Lecture 2		December 24-January 01: Christmas and New Year Vacation
Week 14 Jan 5	Lecture 1	Fundamentals of Digital Logic and Microcomputer Design	January 06: Armenian Christmas (TTh sections) Chapter 5: Sequential Logic Design 5.11 Examples of Synchronous Sequential Circuits: Modulo-n Counters
	Lecture 2		Revision Project submission

III. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	B+
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	
81	3.1		89	3.9	
82	3.2		>=90	4	A

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

XII. Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to

	Program Outcomes
category-1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories, and mathematical methods relevant to computing, computing equipment, computer communication, and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software, and their practical application</i>
category-2 Analysis	PLO -2 Select and use relevant analytic, modelling, and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms, and data structures
category-3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software that meet specified requirements</i>
category-4 Economic, legal, social, ethical, and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring, and use of informatics-based systems
category-5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment, and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6 for CSIT=> Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category-6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO-9 Participate effectively in informatics group work